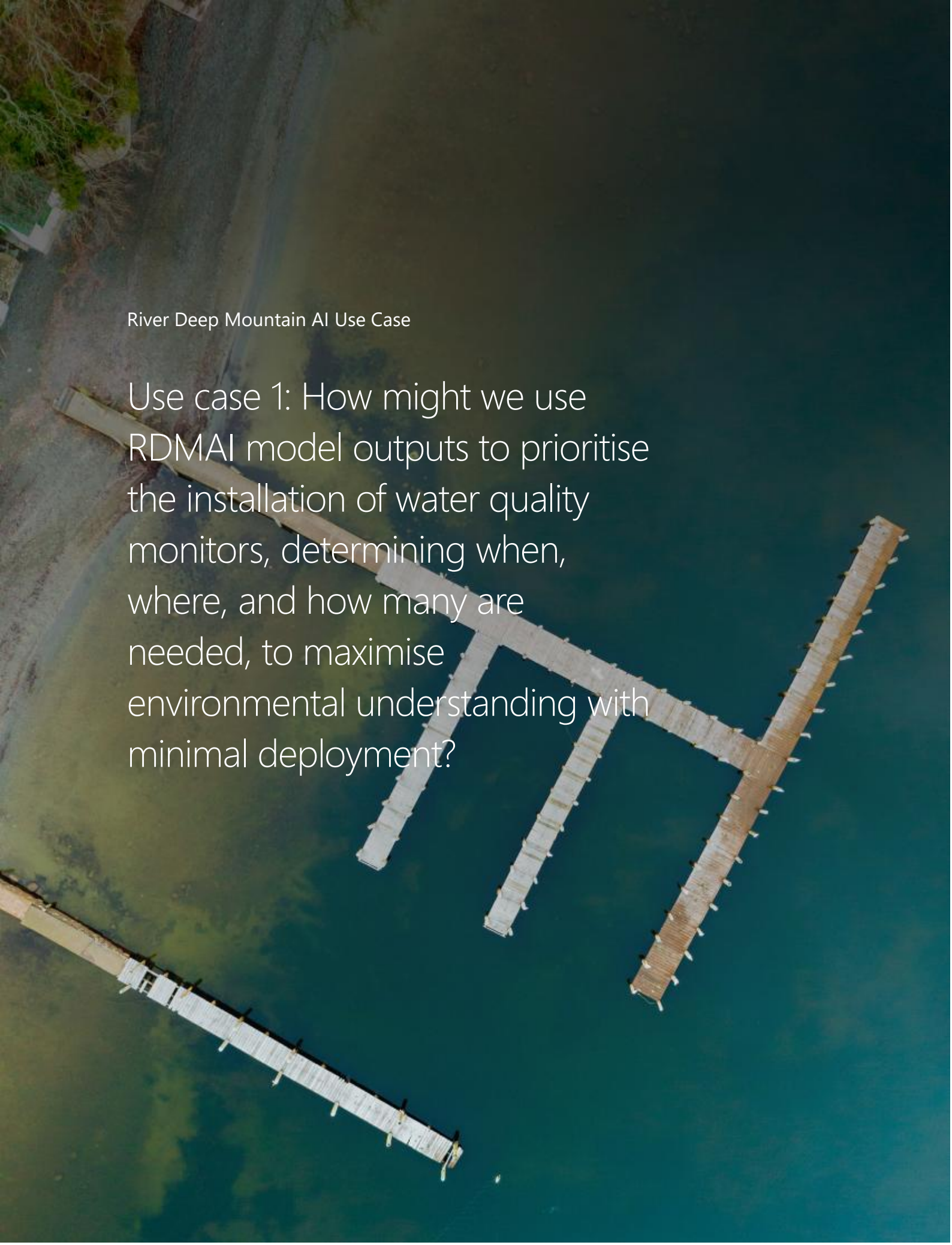

River Deep Mountain AI Use Case

Use case 1: How might we use RDMAI model outputs to prioritise the installation of water quality monitors, determining when, where, and how many are needed, to maximise environmental understanding with minimal deployment?

River Deep Mountain AI

Version overview

Version	Date	Authors	Comments
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Approval

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Executive Summary

This report outlines a use case demonstrating how the River Deep Mountain AI (RDMAI) model suite, specifically the Open Flow, Open Risk Map, and Open Anomaly Detection models, can be applied to strategically prioritise the deployment of water quality monitors. By transitioning from an asset-centric approach to a model-led workflow, catchment managers can maximise environmental understanding while minimising capital expenditure and operational carbon.

The demand for environmental insight is at an all-time high, driven by stricter regulations (e.g., Section 82 of the Environment Act 2021) and public scrutiny. However, the 'Cunliffe Report' (Independent Water Commission, 2025) concludes that a blanket approach of installing hardware everywhere is financially unviable and environmentally unsustainable. Current monitoring is often heavily biased towards water company assets, leaving critical sources of diffuse pollution, such as agricultural runoff, largely unmonitored.

This use case presents an evidence-based workflow to determine where to monitor:

Enhancing the hydrological baseline: The workflow begins by deploying the RDMAI Open Flow Model to generate flow statistics (e.g. mean daily flow). This replaces rudimentary length-based proxies in the Open RDMAI Risk Map, to reduce the risk of flow being underestimated in lower-volume tributaries.

Spatial triage: The RDMAI Open Risk Map ingests this refined flow data and identifies high-risk nodes for Ammonia and Orthophosphate. Users can isolate specific hotspots (e.g., the top 10th percentile of risk) to target investment where it is most needed.

Network optimisation: To prevent redundancy, the workflow compares potential pollution issues detected in sonde data between adjacent nodes, using the RDMAI Open Anomaly Detection Model.

The workflow is demonstrated with two examples: the River Browney catchment in County Durham, a focus area for the *Thriving Catchments* partnership between Northumbrian Water and The Rivers Trust (section 5), and the Rivers Chelmer and Blackwater, a focus for Essex & Suffolk Water drinking water quality investigations (section 6).

At the outset of the RDMAI project, interviews were conducted with water industry stakeholders, aiming to understand how AI and machine learning (ML) models could assist with managing water quality. At the interviews, the stakeholders explained how decisions on water quality monitoring and investments are informed by government strategic policy statements, Water Industry National Environment Programme (WINEP) driver guidance, Water Framework Directive (WFD) status assessments, protected area requirements and local catchment evidence and hydrological models.

The stakeholders explained that being able to compile evidence packs for regulatory bodies (e.g. Defra and the Environment Agency) was essential to show where pollution is coming from and make the case for targeted, cost-effective measurements. As such, monitoring decisions are highly evidence driven but can be constrained by regulatory rules on acceptable methods, regulatory interpretation of environmental data and the need for simple explanations for non-technical audiences.

This use case demonstrates how the RDMAI models could be applied to inform monitoring decisions, allowing the water industry to more effectively build evidence packs, while communicating to regulators and other stakeholders with clarity. To overcome technical barriers, the report recommends approaches for model adoption. The use case finally considers next steps and opportunities for further model development.

1 Prioritising the deployment of water quality monitors in rivers

Water companies, catchment managers, and other stakeholders face significant time and financial outgoings when attempting to design and implement water quality monitoring schemes within a catchment. Furthermore, it can be challenging to determine which sources of pollution represent the highest risk at different locations within a catchment, without prior water quality monitoring.

Water quality monitoring is an essential tool for understanding water body health. There are a range of motivations for monitoring.

1.1 Statutory and national strategic monitoring

Water quality monitoring can be determined by regulatory requirement. For example, Section 82 of the Environment Act 2021 gives limited discretion on the placement of water quality monitoring. The legislation has high obligations for coverage, using an asset-centric approach and a requirement for monitoring to be upstream and downstream of wastewater treatment works outfalls and Storm Overflow (SO) discharge points. A single water company may need to deploy thousands of monitoring pairs across their entire region to cover the relevant assets, regardless of whether a specific river reach is currently at risk of poor water quality.

As a further example, the Water Framework Directive (WFD) (UK Government, 2017) sets out three categories for monitoring, which have different but complimentary purposes: surveillance, operational, and investigative (SEPA, 2007). The categories inform the deployment of sensors and spot sampling, spatially and temporally. Surveillance aims to create a geographically distributed network, designed to allow for monitoring long-term changes. The operational component is driven by risk assessments of catchment pressures, with monitoring then located in areas of known

risk. This monitoring is also designed to assess any changes in the status of water bodies resulting from catchment interventions, potentially inform the reasons for not achieving good (RNAG), and inform WFD river basin management plans. The investigative component is a more variable network, responsive to unplanned events and emerging risks. This monitoring is carried out for any exceedances and to ascertain the magnitude and impacts of accidental pollution.

The Environment Agency, through the Natural Capital and Ecosystem Assessment (NCEA) programme, has recently developed new monitoring strategies, such as the River Surveillance Network. This network is designed to be statistically representative but spatially random, allowing monitoring of biological and chemical metrics of river health to create a robust national baseline (Defra, 2025).

1.2 Monitoring to address local priorities

The timing and location of monitoring may be determined by specific catchment activities, for example the presence of bathing waters or abstraction points. The Environment Agency conduct weekly monitoring of bathing waters from May to September, allowing estimates of microbial water quality to be made, and informing the public about bathing water quality. During the winter season, this monitoring ceases or reduces, as the risk reduces with fewer people using the bathing water (UK Government, 2013).

The water industry may also employ an operational approach, targeting monitoring resources in locations where previous issues have occurred, or where water quality breaches could lead to fines or reputational damage. Monitoring locations may be determined by WINEP (Water Industry National Environment Programme) investigation commitments (UK Government, 2022). These investigations can be designed to assess the potential impact of storm overflows or wastewater treatment works on receiving waters, often planned in accordance with the Urban Pollution Manual (The Foundation for Water Research, 2018). Investigations are also planned to investigate drinking water supply quality risks (see section 6 for example). If a pollution incident occurs in a previously unmonitored watercourse, a sensor might be deployed post-event to monitor recovery and prevent recurrence.

Water quality monitoring location decisions may also be led by citizen scientists, where local concern or engagement in environmental protection can be a driver for monitoring. Activities from stakeholders like The Rivers Trust and local catchment partnerships can play a crucial role in eliminating gaps in monitoring areas that are not the subject of regulatory or industry monitoring. Citizen scientists have become a powerful driver of innovation, evolving from dip-sampling to deploying tools like smartphones and wildlife cameras. Projects such as 'Eyes on the Water' exemplify this shift, demonstrating how accessible technology can expand both the capabilities of volunteers and the geographic scope of monitoring networks. Northumbrian Water

has pledged to double the number of citizen scientist 'Water Rangers' that it supports (A vision for our coasts and rivers 2024, pledge 9).

1.3 Monitoring to inform water quality modelling

Water quality monitoring is critical for water quality modelling as observed data are essential for calibrating and validating a model. Within water companies, modelling may be undertaken to understand the impacts of wastewater assets on river health, or to understand risks to the quality of drinking water supplies. While routine monitoring can be spatially and temporally sparse, limiting effective model testing, specific regulatory investigations necessitate more comprehensive datasets. For instance, studies conducted under the Urban Pollution Manual can require a full year of sonde data supplemented by targeted sampling (e.g. 3 wet and 2 dry events) to capture dynamic conditions.

As regulatory requirements tighten around SOs, reliance on water quality models such as SIMPOL ICM increases the demand for water quality monitoring data. The demand is driven by the Storm Overflow Assessment Framework (SOAF), a structured approach designed to assess the performance of storm overflows and prioritise improvements based on their environmental impact and spill frequency. The SOAF guidance has evolved:

- SOAF Phase 1 (Water UK and Environment Agency, 2018): Established the initial risk-based approach, prioritising investigations for overflows spilling frequently (typically triggered at 40 spills per year).
- SOAF v2 (Environment Agency, 2025): The revised guidance aligns with the Storm Overflow Discharge Reduction Plan, introducing stricter spill frequency triggers (reducing investigation thresholds to as low as 10 spills per year) and requiring clear pathways for improvement.

1.4 Cunliffe Report

The Independent Water Commission's Final Report (the 'Cunliffe Report') details how the demand for environmental insight is at an all-time high, but the traditional resources to deliver it are critically constrained (Independent Water Commission, 2025). The Commission acknowledges that while WFD monitoring budgets are stretched, the water industry possesses the potential to bridge this gap, specifically regarding the impacts of sewage pollution. However, the report is careful to dismantle the assumption that simply installing more hardware is a silver bullet.

Crucially, the Commission found that a 'blanket approach', attempting to roll out continuous water quality monitors to all sites, is fundamentally unviable. The costs of such a strategy would significantly exceed the Environment Agency's entire water quality monitoring budget. Beyond the financial barrier, the report criticises this hardware-heavy approach for being carbon-intensive and operationally rigid, noting that fixed sensors often measure limited parameters and are difficult to adapt for

future requirements. Furthermore, a reliance on fixed infrastructure risks creating a 'data bias' towards urban environments, leaving critical diffuse pollution sources, such as agricultural run-off in rural catchments, largely unmonitored.

Consequently, the Commission argues that the sector must pivot towards smarter efficiency rather than just more hardware. Evidence presented to the Commission suggests that alternative approaches, specifically better data integration and the mobilisation of citizen science, offer a pathway to achieve greater coverage without the unsustainable price tag. With regulator resources reduced even as responsibilities increase, the report concludes that monitoring must be adequately resourced, but those resources must be deployed with far greater strategic precision.

The report also recommends:

- Future water monitoring programmes must be reviewed and adequately resourced to ensure they provide a comprehensive and accurate reflection of the state of the water environment.
- The current system of Operator Self-Monitoring should be significantly reformed to include greater digitisation, automation, third-party assurance, and public transparency to rebuild trust.
- The government should conduct a review of the Continuous Water Quality Monitoring programme to evaluate its effectiveness and value for money, seeking cost-efficiencies through new technology.
- Monitoring should be modernised by accelerating digital programmes for intelligence-led compliance and better integrating diverse data sources, including citizen science, to extend coverage.

2 The challenges of prioritising monitoring locations

One of the challenges in water quality management is financing a network of sensors sufficient to adequately understand water body health. Traditionally, each UK regulator maintains a routine water quality monitoring programme alongside a network of continuous monitoring stations. This monitoring is often strategically concentrated at river confluences or at the bottom of catchments. While this approach allows for an understanding of the overall, cumulative impacts on water quality, providing a basis for surveillance of long-term trends and model validation, it leaves significant gaps in our understanding of the upstream system.

Specifically, monitoring at the downstream end of a catchment often lacks the spatial resolution required for effective source apportionment. Without granular upstream data, it is difficult to identify the relative contributions of specific pollution sources, such as distinguishing between a continuous discharge and diffuse agricultural runoff. Furthermore, source apportionment is not just a spatial challenge; it has a

temporal dimension. Many pollution sources are event-based, driven by rainfall, runoff, or SO spills, meaning they are easily missed by routine spot sampling. Consequently, the challenge of prioritisation is two-fold: stakeholders must determine not only where monitoring should take place to isolate sources, but also what kind of monitoring (spot samples vs. high-frequency sensors) is required to capture the full temporal dynamic of pollution events. Some pollutants cannot be measured using sensors, therefore samples analysed in a laboratory are needed.

While the implementation of Section 82 of the Environment Act 2021 is set to drive a massive expansion in sensor deployment, this monitoring is wastewater-asset centric. The legislation mandates the monitoring of water quality upstream and downstream of storm overflows and wastewater treatment works (UK Government, 2021), meaning the resulting data will be heavily biased towards water company infrastructure. Consequently, while this will provide unprecedented insight into sewage impacts, it leaves significant blind spots regarding non-water company outfalls, such as highway drains, agricultural runoff points and private discharges, which remain largely unmonitored despite their contribution to catchment pollution.

3 Using the RDMAI models to prioritise the installation of water quality monitors

To help stakeholders deploy ammonia and orthophosphate monitoring more effectively, and address the challenges identified in the Cunliffe report, this use case demonstrates how three of the eight open-source RDMAI models can inform the prioritisation of water quality monitoring locations. Based on the current model set up, the use case could be applied to a catchment in England. Stakeholders could subsequently use the monitoring data to inform evidence packs for regulatory bodies.

Effective prioritisation of monitoring locations requires viewing pollution sources and hydrological context in parallel. Consequently, the RDMAI Open Risk Map and Open Flow model were selected to provide this dual perspective. The Risk Map defines the spatial 'where' by identifying high-risk locations along a river, while the Flow Model could improve the accuracy of the Risk Map and help characterise the risk.

Finally, the use case extends into the operational phase, demonstrating how the RDMAI models can be applied to detect anomalies in continuous monitoring data, thereby aiding stakeholders in the complex task of interpreting the data generated by continuous water quality monitors.

Further details on each of the models discussed can be found in their respective model documentation.

3.1 The Open Risk Map

The Open Risk Map is built up of individual risk layers, each of which aligns with a particular risk type (orthophosphate-P and ammoniacal-N) and source pressure informed by the EAs reasons for not achieving good (RNAG) status, including:

- Sewage discharge (continuous), i.e., Wastewater Treatment Works (WwTWs) final effluent discharges
- Sewage discharge (intermittent), i.e., overflow spills
- Agricultural runoff
- Urban runoff
- Private sewage treatment.

Each risk layer uses openly available data to calculate a concentration, which is added to a baseline catchment map (Figure 1). The Open Risk Map then determines how the level of risk varies throughout the river network in a catchment. Risk Map outputs compare reasonably well against current industry-standard modelling software (SAGIS) – as shown in the model benchmarking reports available here.

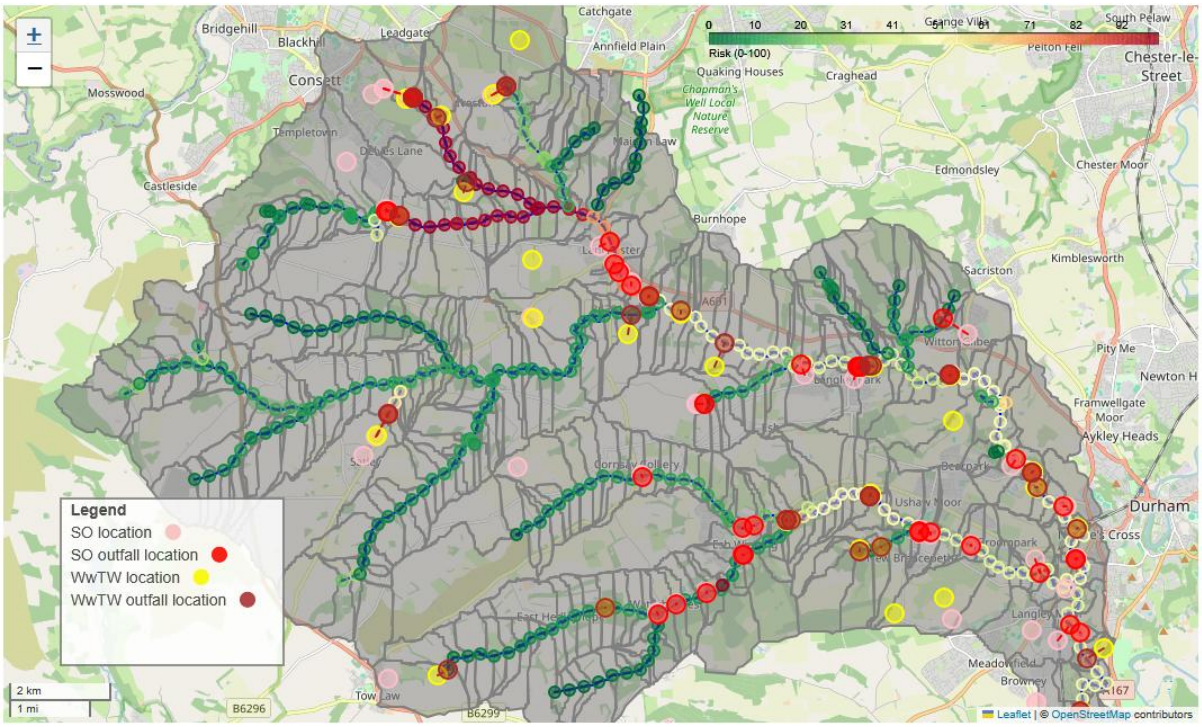

Figure 1 Risk distribution for the River Browney catchment, County Durham. The output is shown in a Jupyter Notebook.

3.2 The Open Flow model

The RDMAI Open Flow model is a machine learning model designed to estimate daily mean river flow (m^3/sec) in rivers. Built on a Long-Short-Term Memory (LSTM) architecture within a recurrent neural network (RNN), the model integrates static (catchment attributes which rarely change) and dynamic data sources, including historical flow, weather, and catchment attributes from the CAMELS-GB dataset, the

Environment Agency's Hydrology Data Explorer, and a curated list of chalk streams from Natural England.

The CAMELS-GB dataset spans 40 years of daily river flow records from 409 catchments across England, enabling the model to learn complex relationships between weather patterns, topography, and catchment characteristics. Dynamic time-series data are processed through an LSTM layer using a 30-day window to capture antecedent effects on flow, while static attributes are handled separately. The outputs from these two branches are then combined to produce a single flow prediction for each catchment (Figure 2).

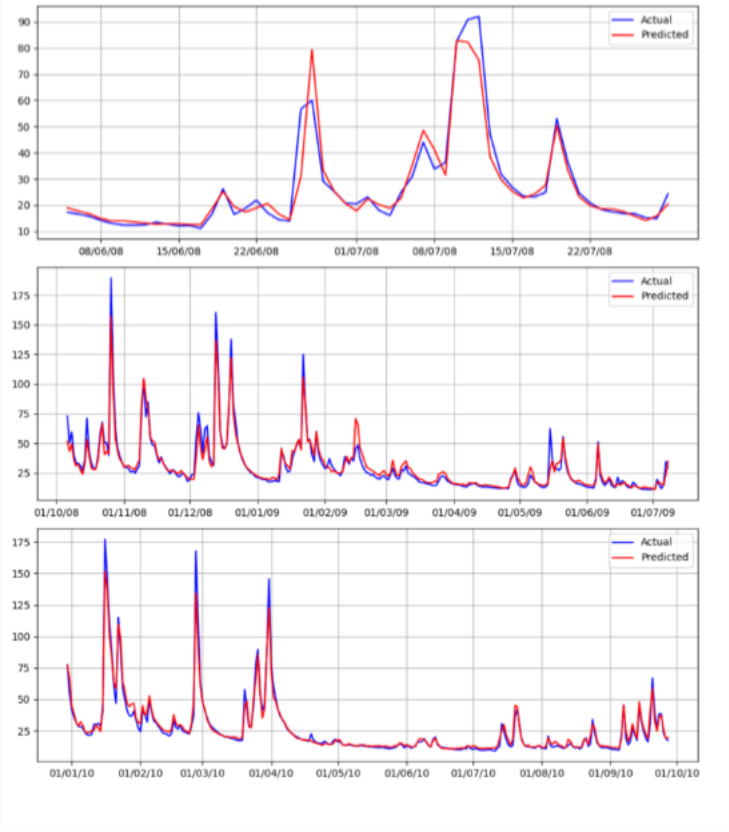

Figure 2 Example of output from the Open Flow model for the Aire at Beal (Gauge: 27003)

3.3 The Anomaly Detection model

The Anomaly Detection model presents an end-to-end (E2E) framework designed to ingest, process, and analyse multi-parameter water quality data for UK water utilities, specifically focusing on the Section 82 parameters: pH, dissolved oxygen (DO), turbidity, temperature and ammonia. The model also considers conductivity.

The workflow begins with data cleaning and standardisation, followed by a decomposition stage where seasonal and trend components are extracted. Once these underlying patterns are removed, the residual signals are analysed using an Isolation Forest unsupervised machine learning algorithm to detect anomalies, which are subsequently grouped into distinct clusters using KMeans.

While the model is a powerful tool for exploring data behaviour, automating trend analysis, and highlighting potential irregularities, it does not definitively classify the type of anomaly detected. Consequently, the framework is intended as an exploratory guide, and its results regarding water quality changes should be interpreted with caution and validated by domain knowledge. Data irregularities flagged by the model, indicating potential pollution events, could be investigated further by domain specialists.

3.4 Data sources required

To run the three models, various data sources are required. All data used in the models are open source and free to download. As the RDMAI models are open source, the user could amend the models to enable the use of alternative data. This use case focuses on the data sources used in the development of each model.

RDMAI Model	Data Sources
Open Risk Map	The EA consents database needs to be manually downloaded into a stipulated folder. All other datasets are processed automatically by the provided script. • Rivers: Open River Network • Overflow spill data: EA Storm Overflows - Annual Returns • Consents data: EA Consents Database • Land use: CROME 2023 dataset • Hydrology of Soil Types (HOST) derived from this study. • Rainfall data: UK Met Office Rainfall Data • Roads and buildings: OS OpenMap - Local • Built-up areas: OS Open Built Up Areas • Farmscoper Data: ADAS Farmscoper Tool is used for agricultural and livestock runoff estimates via lookup tables.
Open Flow Model	• Catchment attributes: CAMELS-GB • EA hydrology data: Hydrology Data API • Natural England chalk stream locations: Chalk streams in England:
Open Anomaly Detection Model	• CSV file containing all six expected water quality variables (DO, pH, turbidity, ammonia, conductivity and temperature)

4 Workflow – using the RDMAI models to prioritise the installation of water quality monitors

4.1 Model set up and stakeholder requirements

Users will need to follow the instructions provided on the model GitHub repositories to download and run the models.

4.1.1 Open Flow Model

The RDMAI Open Flow model involves a relatively complex setup process. Installing the model and its dependencies, downloading and preparing the necessary datasets, and running the trained model for a prepared example generally takes 1 to 2 hours, providing the catchment is included in the CAMELS-GB dataset. Most of this time is spent on preparing the catchment datasets, while executing the trained model itself takes only a few seconds once data preparation is complete. The stakeholder could train the model to their specific catchment, but this would involve additional time and expertise. Due to these requirements, the Flow model is best suited for stakeholders with intermediate Python skills and familiarity with scripting and data handling. While the initial setup is time-intensive, subsequent runs are extremely fast, making it efficient for repeated use.

4.1.2 Open Risk Map Model

The RDMAI Open Risk Map model is designed to be highly accessible, with both the model and its input data freely available. Its setup is straightforward and requires basic Python skills, such as cloning a GitHub repository, configuring a Conda environment, and working with Jupyter Notebooks. The model is largely "plug-and-play", enabling stakeholders to run analyses with minimal technical experience. Once the environment is prepared, running a catchment typically takes between 5 and 30 minutes on a standard laptop, depending on the catchment size. This makes the Risk Map model well-suited for users with limited coding experience who need quick and reliable outputs, that are repeatable for multiple catchments (Figure 1).

4.1.3 Anomaly Detection Model

Monitoring programmes to understand the impacts of wastewater assets on receiving waters often include sondes measuring dissolved oxygen, pH, temperature, conductivity, ammonia and turbidity. When these are installed, the RDMAI Open Anomaly Detection Model can be run to identify 'spikes' indicating potential pollution events. Installing and running the RDMAI Anomaly Detection Model requires a basic knowledge of Python and will take between 1 and 2 hours. Once the model is installed, running the anomaly detection model on around 2 months of data takes seconds.

4.2 Defining the monitoring strategy

4.2.1 Enhancing the hydrological baseline

The workflow begins by improving the flow estimate used in the Open Risk Map. In its standard configuration, the Risk Map uses a simple proxy for estimating flow, relying on upstream accumulated river length to estimate flow. While this allows the model to run rapidly without complex data, it introduces uncertainty, as it does not account for the differences in flow between different waterbody types (e.g. a chalk stream versus an upland stream).

To help overcome this limitation, in this use case workflow, the user first deploys the RDMAI Open Flow model to generate flow statistics for a catchment. At present, the catchment would need to be included in the CAMELS-GB dataset and modifications to the Risk Map model code would be required. Although the Risk Map does not ingest daily time-series data, the Flow Model is used to generate aggregate statistics, such as the mean daily flow or the median flow (Q50). These flow values could then be substituted into the Risk Map's input parameters, replacing the rudimentary length-based proxy. This step could improve on the Risk Map's dilution calculations, preventing the model from underestimating risk in tributaries where the flow volume is lower than the length would suggest. In catchments that are not part of the CAMELS-GB dataset, the user could still use the default flow estimation method within the Risk Map.

4.2.2 Where to deploy

With the hydrological inputs refined, the user runs the Open Risk Map to perform a catchment-wide triage. The model functions as a quantitative screening tool, ingesting the refined flow data alongside land use and point-source data to map pollutant risk across the entire river network. The results are visualised directly within the Jupyter Notebook interface, where nodes are colour-coded from green to red based on the intensity of the predicted risk for Orthophosphate-P or Ammoniacal-N. The user may wish to export the results as a GIS shapefile for further analysis.

Once in a GIS environment, the user can filter these results to identify specific hotspots where risk values for Orthophosphate-P or Ammoniacal-N are greatest. This filtering process can be tailored to the available resources; for instance, a user might isolate the top twenty highest-risk nodes to match a specific hardware budget or target the top 10th percentile of risk.

Crucially, the model allows the user to inspect the relative contribution breakdown at each node. This diagnostic capability can inform the right monitoring equipment and methodology; if the breakdown reveals the primary pressure is continuous (such as final effluent from a Wastewater Treatment Works), the user might prioritise constant physico-chemical monitoring. Conversely, if the risk is driven by urban runoff, the

user must be aware that high flows may carry pollutant spikes rather than providing dilution, suggesting a need for event-triggered sampling.

4.2.3 How many monitors to deploy

Interpreting and analysing the vast quantities of multi-parameter data poses a challenge. To assist with this, catchment managers could deploy the Anomaly Detection model to process the incoming data streams. This tool offers a potential method for network optimisation, by comparing the anomaly signatures between adjacent monitors, allowing the user to assess if they are recording unique events or duplicating data. If similar anomalies are consistently detected at both locations (e.g. a turbidity spike traveling downstream), it may suggest that the reach is uniform, prompting decision makers to review and decide whether one sensor could be removed. This would support an adaptive monitoring plan, where the network layout evolves based on the observed data, ensuring resources are continuously redeployed to where they add the most unique value.

4.3 Adoption

As discussed previously, monitoring locations may be determined by legislative requirements or to address local concerns regarding water quality. Successfully transitioning from existing monitoring prioritisation to the RDMAI workflow requires a shift in both technical capability and operational culture (Figure 3). A significant barrier to immediate adoption is the technical disparity between the plug-and-play Risk Map and the more code-intensive Flow Model. Expecting every Catchment Manager to become proficient in Python is unrealistic and could stall implementation.

To overcome this, the organisation could adopt a 'Hub-and-Spoke' operational model. In this structure, a designated Data Scientist acts as the 'Hub', responsible for code management, data curation, and model execution. The Catchment Managers then act as the 'Spokes', receiving the processed outputs, such as GIS shapefiles and flow CSVs, and applying their domain expertise to interpret the results. This workflow ensures that high-level modelling is accessible to decision-makers without requiring them to learn coding.

Adoption also necessitates a re-engineering of business processes. Currently, site selection may be reactive, driven by public complaints or regulatory requirement, or based on historic precedent where sites are monitored simply because they always have been. The RDMAI workflow could replace this with a proactive, evidence-based standard. To drive adoption, the business could mandate that requests for new monitoring capital expenditure must be accompanied by RDMAI model outputs.

While the current code-based format is powerful, widespread democratisation of these tools will eventually require a user interface. Because the RDMAI project is open source, organisations have the flexibility to develop an internal project,

wrapping the underlying Python code into a simple web-based interface. Developing a self-service interface would allow operational staff to select a river reach on a map and run the analysis independently, removing the need for input from the Data Scientist. The RDMAI project team has submitted a proposal to undertake this work. Contributing these improvements back to the open-source community allows the water company to benefit from updates from other stakeholders, ensuring the tools remain cutting edge.

An organisation could also take the opportunity to refine the models for use in their catchments.

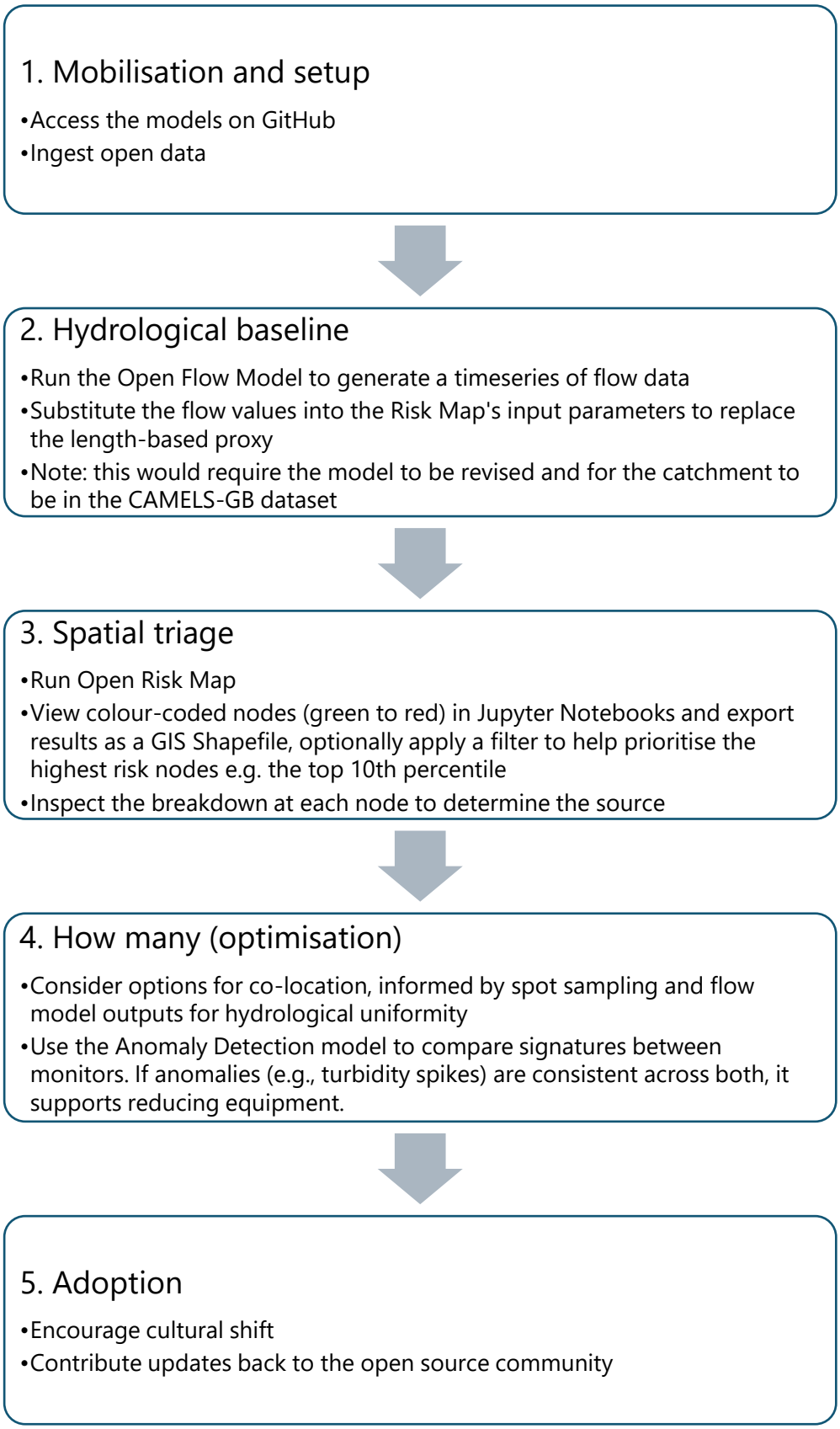

Figure 3 Example workflow for prioritising monitoring locations using the RDMAI models

4.4 Value propositions

Beyond the technical optimisation of the monitoring network, integrating the RDMAI models into catchment planning delivers tangible business value across financial, regulatory, and environmental metrics.

4.4.1 Financial efficiency

With an estimated 28,000 monitors required across AMPs 8 and 9 for Section 82 compliance, carrying an estimated capital cost of £15,000 each for installation, the water industry faces an estimated £420 million capital expenditure. The shift to a model-led workflow helps prevent over-specification and unnecessary deployment of monitoring equipment, offering a pathway to help mitigate this financial burden

By shifting from a rigid, blanket rollout to an intelligence-led deployment, catchment managers can ensure fewer monitors are used, but that they are more intelligently placed.

Furthermore, redundancy checks (using the Anomaly Detection models) would eliminate duplicate monitors, saving money. Securing budget for monitoring hardware is often competitive against other business needs. Instead of relying on intuition, a catchment or asset manager can present a business case backed by evidence-based model outputs.

The model benchmarking reports provide further information on the cost and time savings that could be expected by using RDMAI models over existing industry tools and models.

4.4.2 Protecting business metrics

For water companies, this workflow directly supports performance against performance commitments set by regulators.

4.4.2.1 Pollution incidents (Category 1-3)

The Flow Model identifies low-dilution windows, which may increase the likelihood of a Category 1-3 pollution incident. By proactively monitoring high-risk sites, water companies can detect potential incidents, allowing for intervention that prevents a pollution event.

4.4.2.2 WINEP compliance

The Risk Map could act as a root cause analysis tool, providing evidence on whether a failing reach is due to water company assets or diffuse agricultural pollution. This ensures capital investment is directed only towards assets where it will yield regulatory compliance, avoiding expenditure on problems outside the company's control. The Risk Map has potential to inform WINEP drinking water quality investigation commitments, as illustrated by the example in section 6.

4.4.3 Strategic benefits

4.4.3.1 Carbon reduction

A streamlined network requires fewer physical assets, reducing embedded carbon. The optimised deployment locations reduce field visits, reducing fleet mileage and contribute to Net Zero goals.

4.4.3.2 Regulatory trust

As evidenced in the Cunliffe report, regulators (such as the Environment Agency) are increasingly looking for risk-based approaches to catchment management. Demonstrating that the monitoring network was designed using a rigorous, open-source multi-model methodology (screening for load, validating for flow, and optimising for redundancy) provides a strong audit trail that demonstrates due diligence and responsible asset management.

4.4.3.3 What-If scenario testing

The models allow for proactive rather than reactive planning. A manager can simulate future scenarios, such as the addition of a new housing development (increasing risk load) or the impact of climate change (altering flow regimes), to anticipate where *future* monitors will be needed.

4.5 Limitations

While the use case workflow offers a tool for evidence-based planning of monitoring locations, users must be aware of specific constraints within the RDMAI models.

The Open Risk Map is designed to provide an annual average estimation of risk across a catchment. Consequently, it does not intrinsically represent seasonal variations, rainfall-driven pollution spikes, or acute low-flow events, except insofar as these factors influence the annual average. Furthermore, the model relies on a rudimentary approximation of river flow to estimate in-stream pollutant dilution. Validation exercises have demonstrated that risk estimations could be considerably improved by substituting this approximation with measured flow data.

There are also limitations regarding the source data used to calculate pollutant loads in the Risk Map. Estimates for point source inputs, such as WwTWs and SOs, are derived from open data sources like the consents database. As the discrepancy between consented and actual discharges varies by source type, it impacts the modelled propagation pathways and overall risk calculations. Crucially, this uncertainty is likely to be inconsistent across all pollution sources, and therefore could lead to a systematic under-representation of the impact from certain sources relative to others. However, this uncertainty is not an intrinsic flaw of the model and can be reduced by substituting open data with granular, proprietary water company data where available. Similarly, the estimation of agricultural diffuse pollution assumes a uniform distribution of livestock across a catchment, which may oversimplify the complex spatial heterogeneity of farming practices.

While the RDMAI Open Flow Model serves as a check on hydrological dynamics, it is not universally applicable. Presently, flow can only be simulated at CAMELS-GB catchments, all of which have flow gauges, so a user wishing to simulate flow outside of these catchments would need to amend the model script and undertake additional calculations (e.g., creating the CAMELS-GB data or equivalent for ungauged catchments). Furthermore, the model's ability to extrapolate outside its training data is constrained – additional training is likely to be needed. Validation indicates that while the model provides reasonable estimates (NSE of 0.54 to 0.62) of average daily flow for approximately 80% of catchments assessed, it under-predicted flow dynamics in the remaining minority. Preliminary analysis by the Rivers Trust suggests that the Open Flow Model performs better in some river morphologies than others – therefore accuracy is likely to vary depending on the river being investigated. The specific hydrological or geological reasons for these shortcomings are currently undetermined, meaning the model cannot yet be universally applied to every ungauged catchment without a degree of verification. Model improvements would require advanced Python and GIS skills.

The RDMAI Open Anomaly Detection Model has been designed to identify short-term, high-magnitude events, representing large shifts up or down from a baseline. It does not detect slow changes over time – e.g. sensors deteriorating over time. It requires at least one year of historic data to run. To run in real-time, the model structure would need to be changed, prior to integration into any sort of data streaming infrastructure. The model has not been extensively validated, and further checks are recommended before it is operationalised.

Each model has an accompanying report, discussing how the model was trained, evaluated and the validation approach used. The model reports can be viewed here.

5 Case study: prioritising monitoring locations in the River Browney

Northumbrian Water undertook work to progress catchment solutions for achieving WFD targets for phosphorus between 2023 and 2025, supported by the £3m *Thriving Catchments* project. The project aimed to develop proposals for catchment-based interventions to reduce phosphorus in rivers, as more cost-effective and multi-benefit alternatives to end of pipe treatment at WwTWs, aiming to save customers £40m over the 5 year AMP programme. The project was coordinated by Northumbrian Water, supported byThe Rivers Trust, and engaged local catchment partners. The development of proposals was underpinned by an extensive programme of catchment monitoring in order to develop the catchment data and evidence base and gain a fuller understanding of catchment processes and conditions to inform the right solutions. Both continuous monitoring and sampling were undertaken. Since the monitoring work was undertaken, the regulatory

approach to progress investment in catchment solutions through Catchment Nutrient Balancing has been withdrawn by the Environment Agency. However, the dataset remains an important output of the project which will inform future catchment measures to be delivered or co-funded by catchment partners.

One of the initial focus areas was the Browney Catchment. The monitoring locations that were proposed early in the *Thriving Catchments* programme are shown on Figure 4 (green circles), alongside outputs from the RDMAI Risk Map. While the *Thriving Catchments* programme aimed for broad catchment coverage rather than solely targeting risk hotspots, it is notable that Northumbrian Water's selected monitoring sites generally align with the higher-risk river sections identified by the RDMAI Risk Map. However, users must recognise that the model output represents a transparent first pass to help justify monitoring decisions to stakeholders. Final site selection is inevitably influenced by factors outside the RDMAI models' scope, such as available hardware budgets, site accessibility, and discussions with landowners, necessitating a phase of ground-truthing following the initial analysis.

Using the Browney Catchment as a case study, the RDMAI Open Risk Map was deployed to demonstrate how the workflow can be used as a decision support tool to identify locations with the highest orthophosphate (Figure 4) and ammonia risk (Figure 5). In this instance, the Risk Map was run in its standard configuration, without inputs from the Open Flow Model. As discussed in the adoption section, model outputs were exported to GIS allowing integration with existing spatial layers, and a more accessible way for non-data scientists to interpret the outputs.

Nodes representing the top 10th percentile of risk were isolated, and a 3 km separation filter was applied to prevent data confounding (Figure 6). It is important to note that this '10% rule' is a simplified filter used to illustrate the proof of concept. In a fully operational workflow, this logic could be substantially improved. For example, rather than a simple percentile cut-off, priority nodes could be selected based on the rate of change in risk along a river reach (to identify sudden pollution inputs) or by incorporating rules to ensure every major river confluence is monitored. Users may wish to use a different set of filters to target monitoring locations, for example filtering the nodes with the greatest risk from a particular pollution source e.g., urban areas.

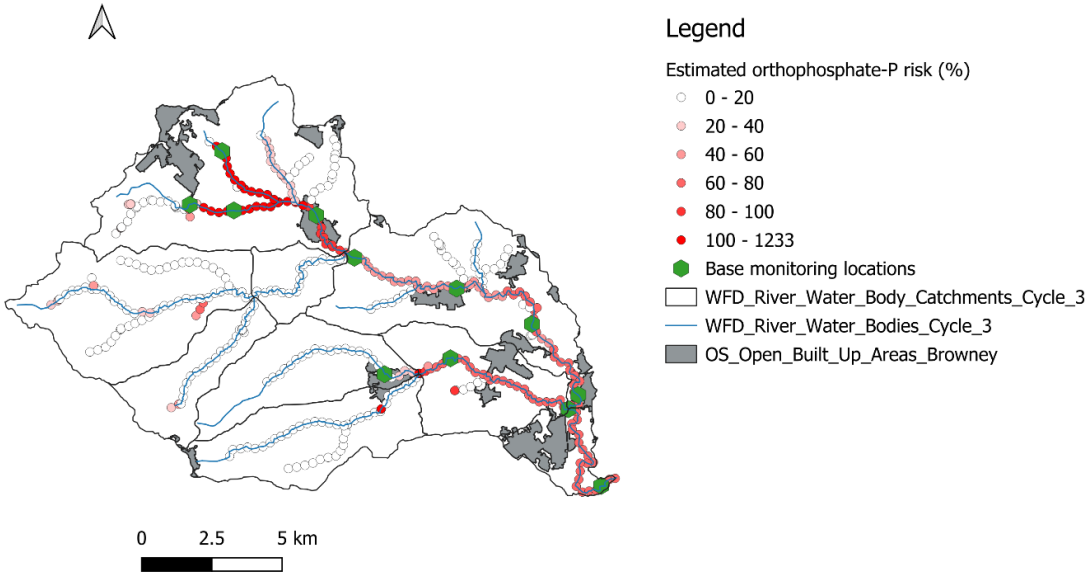

Figure 4 Map of the Browney catchment showing the RDMAI Open Risk Map estimated orthophosphate-P risk [calculated as orthophosphate-P concentration as a percentage of the average of the UK TAG Revised orthophosphate-P standards for poor river health, 0.87 mg/l]. Green circles show the locations initially selected by Northumbrian Water for monitoring/sampling.

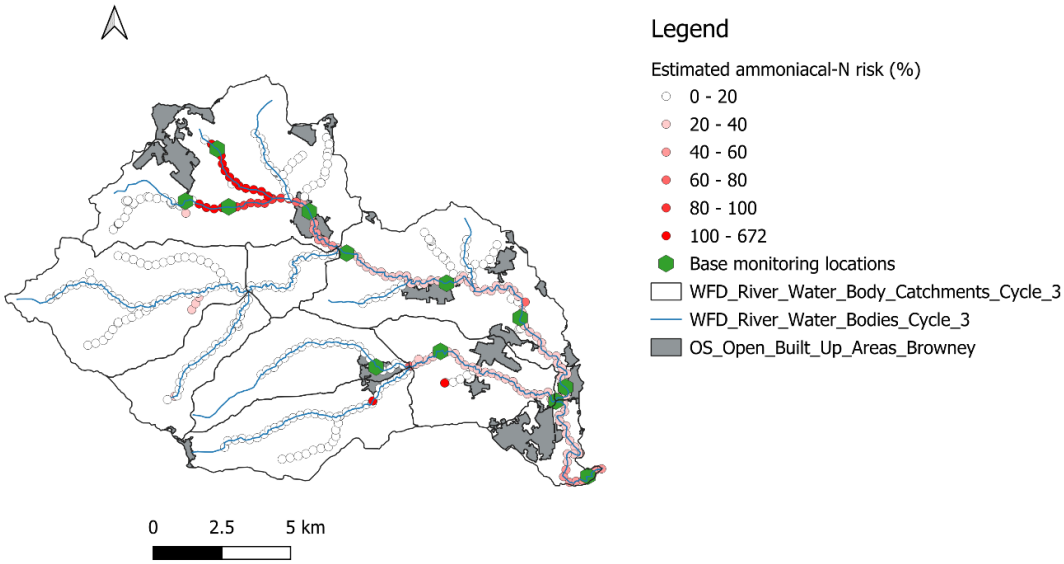

Figure 5 Map of the Browney catchment showing the RDMAI Open Risk Map estimated ammoniacal-N risk [calculated as ammoniacal-N concentration as a percentage of the average WFD 99th percentile standards among the different WFD status categories (Poor category) and and water body types (3, 5 or 7) of 6.0 mg/l].

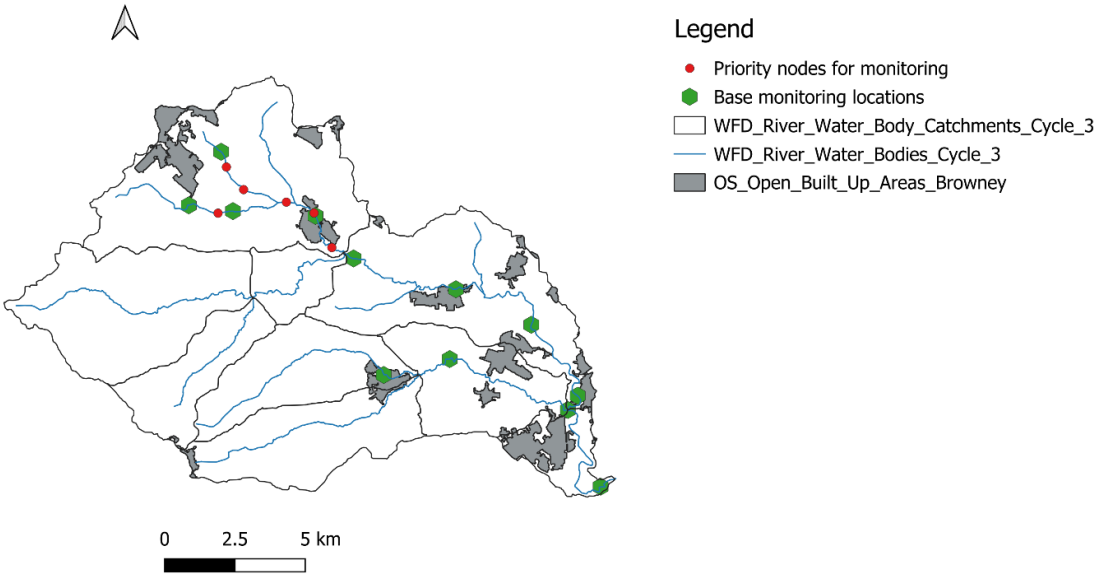

Figure 6 Map of the Browney catchment showing priority nodes for monitoring according to RDMAI Open Risk Map estimated orthophosphate-P risk (Figure 4) and ammoniacal-N risk (Figure 5). Here, the nodes with the top 10% of risk were isolated, and a 3-km separation between nodes to avoid confounding during monitoring.

The user can explore the contribution of different pollution sources, and how each source contributes to overall risk. In this example, at the priority nodes, most of the orthophosphate risk is due to wastewater treatment works (Figure 7) and SOs (Figure 8). Livestock agriculture is the second most prevalent cause (8.2% to 13%), followed by crop agriculture (2.7% to 3.8%).

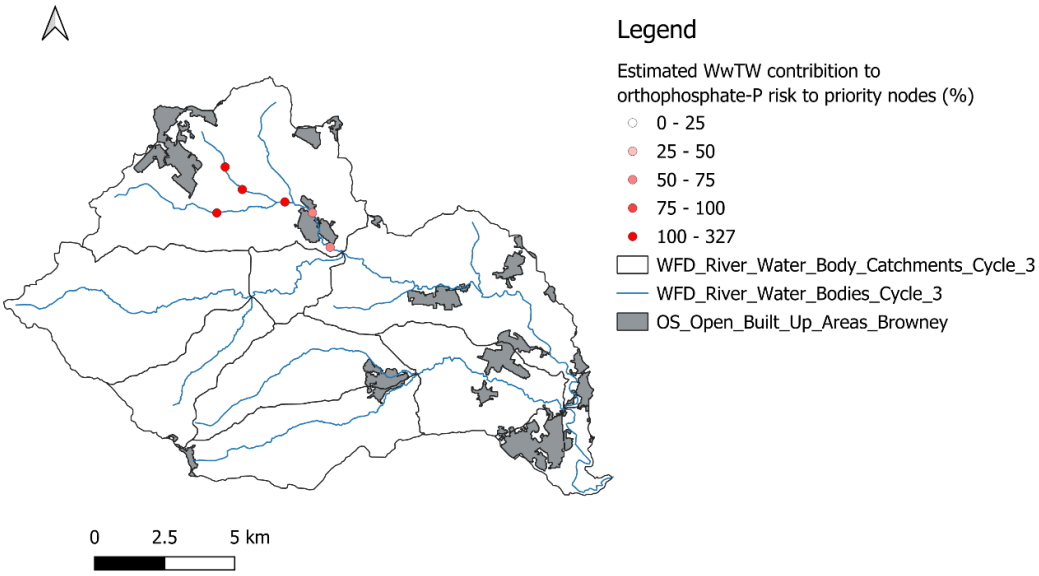

Figure 7 Map of the Browney catchment showing the contribution of wastewater treatment works (WwTW) to orthophosphate-P risk (Figure 1) in priority nodes for monitoring (Figure 5)

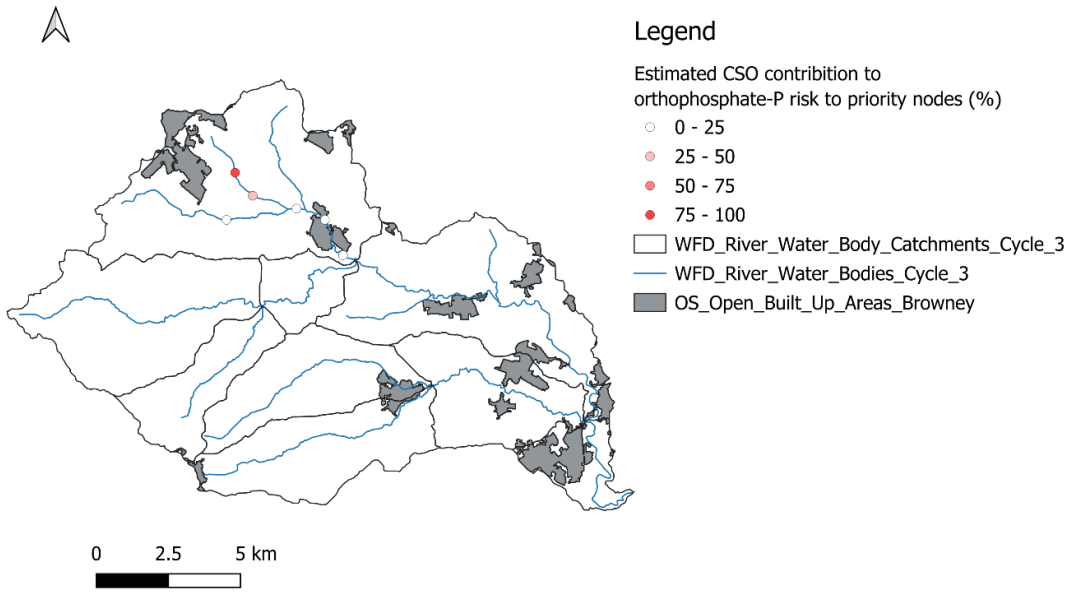

Figure 8 Map of the Browney catchment showing the contribution of storm overflows (SOs) to orthophosphate-P risk (Figure 1) in priority nodes for monitoring (Figure 5)

6 Case study: prioritising monitoring locations in the Rivers Chelmer and Blackwater

Under its PR24 WINEP programme[1], Essex & Suffolk Water (ESW), a water only company under the Northumbrian Water Group, has committed to *'protect and improve water supply quality'* across the Rivers Chelmer and Blackwater, both Drinking Water Protected Areas (DrWPAs)[2]. Substances to be investigated, and - depending on the outcomes of investigations – addressed, include nutrients (e.g., nitrate and phosphate) and algae.. These investigations are part of the ESW WINEP programme of catchment investigations, estimated at £28m, including water quality priorities.

Each WINEP programme commitment has an associated 'Action Specification Form' – approved by the Environment Agency – which outlines the investigation objectives and actions required. These include *'multiple surveys and/or monitoring and/or complex modelling'* to understand the sources; farmer engagement; managing *'fields and boundaries to reduce runoff, sediment and nutrient losses'*; river restoration, and appraisal of other potential mitigation options[1]. When monitoring is undertaken, at least one year of data is generally required to capture seasonal trends, representing a significant cost. Monitoring plans vary but techniques may include continuous

[1] https://environment.data.gov.uk/dataset/39b11ea0-3cfa-4cbb-b3a1-b5950019f169, accessed 20/02/2026

[2] https://environment.data.gov.uk/dataset/b4f0e481-f5e3-4421-90b8-5f1cde58eb59, accessed 20/02/2026

monitoring using sondes and flow monitors, ad-hoc investigations and walk overs, field testing and lab sample analysis, event-triggered storm sampling, and passive samplers.

Monitoring locations and plans are supported by consultants, asked to provide recommendations through undertaking reviews of available data, and identifying gaps in current knowledge. These reviews draw on the Environment Agency's Water Quality Archive (WIMS), along with local data that the water company has gathered.

This is not straightforward. Catchment data are often limited and fragmented. Information on pollutant source apportionment can be lacking, particularly where wastewater infrastructure and water supply infrastructure are managed by different companies - Anglian Water, and Essex & Suffolk Water in this case. At the early stages of planning a monitoring programme, risk may be based on an individual consultant's interpretation of background mapping and likely pollution sources. Outputs from typical water quality modelling tools like SAGIS can be difficult to access, present and discuss with the many stakeholders involved.

Figure 9 illustrates how the RDMAI risk map could help. It quickly (<2h) provided a view of pollutant risks in the Rivers Chelmer and Blackwater, and can be readily understood by stakeholders from within and outside water companies. Suggested monitoring points have been rapidly identified by selecting nodes where the change in risk is greater than or equal to 20%, and ensuring more than 3 km between monitors. This is likely to be more monitors/samples than allowed for in the company budget. By overlaying this map with existing monitoring points from WIMS and Essex & Suffolk Water local data, data gaps could be identified. This would illustrate where there are potential pollution risks that are currently unmonitored. Monitors at headwaters could also be removed as the monitoring plan is refined.

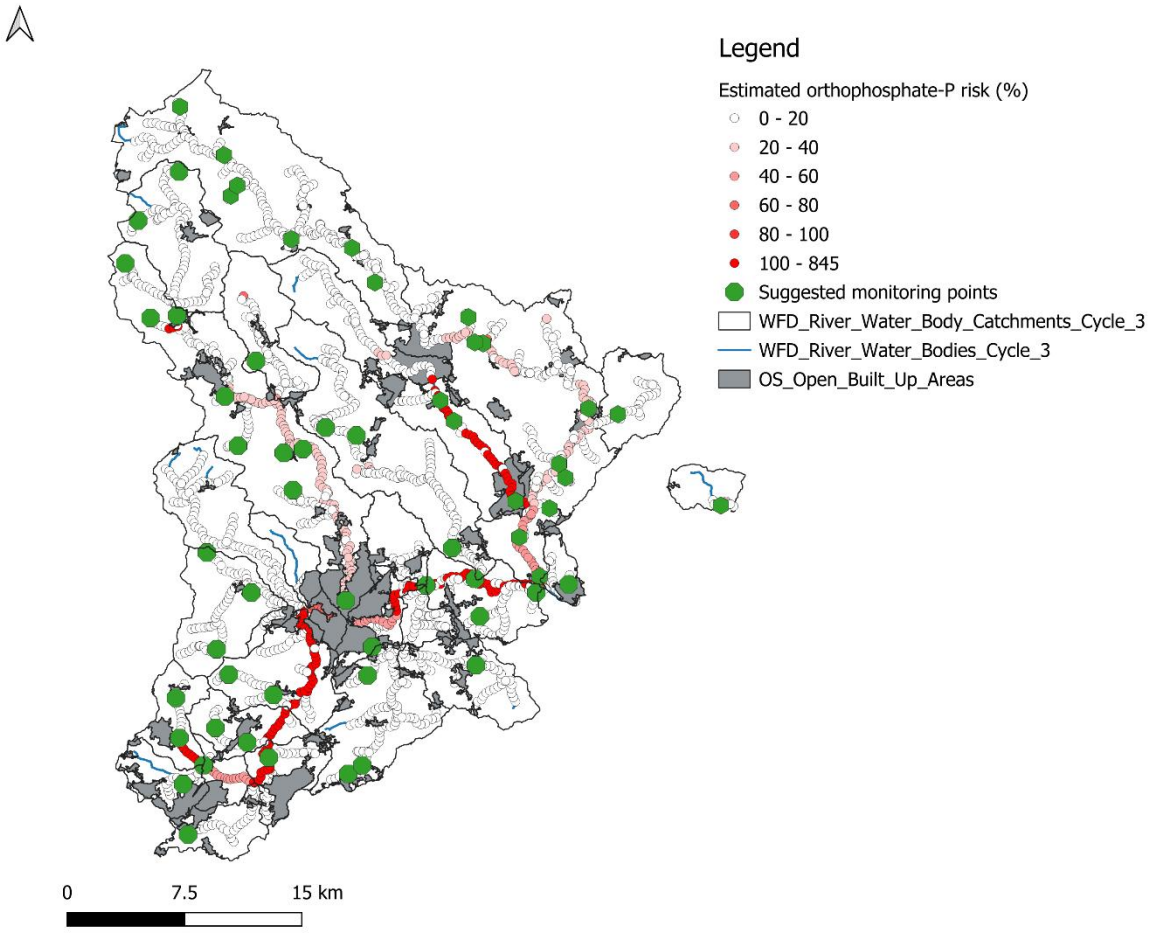

Figure 9 Map of the River Chelmer and Blackwater catchments, showing RDMAI risk map outputs and potential monitoring points. Risk is calculated as orthophosphate-P concentration as a percentage of the average of the UK TAG Revised orthophosphate-P standards for poor river health, 0.87 mg/l.

The current iteration of the RDMAI risk map works for orthophosphate and ammoniacal nitrogen. It would need to be adapted to investigate nitrates, for example. Further model adaptations could include:

- Enabling the map to be run for different periods – such that changes over time could be investigated. Many of the drinking water catchments in Essex & Suffolk Water's PR24 WINEP programme did not have issues five years ago, so understanding changes over time could be significant.
- Adapting the risk map so that it can be applied to lakes/reservoirs, or areas impacted by the tide or groundwater.

The risk map is not a substitute for local knowledge. However, when used alongside local insights and consultant domain expertise, it could enable quicker, more transparent and more robust planning of water quality monitoring programmes.

7 Next steps and opportunities

The workflow described and demonstrated above uses the individual RDMAI models to prioritise monitoring locations. Integrating the individual models into a single platform with a user-friendly interface, would widen the pool of potential end users. Overlaying model outputs with maps showing existing data from regulators, water companies, citizen scientists and satellites would further improve accessibility and usability; the RDMAI team has developed proposals to undertake this work.

A further opportunity for development of the use case lies in transforming the current hindcast workflow, which looks at historical data to plan future actions, into a real-time nowcasting system. By creating an automated pipeline that feeds live meteorological telemetry (e.g. rainfall radar) directly into the Open Flow Model, the tool could evolve into a near-real-time Digital Twin of the catchment. This would allow catchment managers to predict high-risk pollution events days in advance, moving the use case from a strategic planning tool to an operational warning system. Furthermore, addressing the usability gap would improve access, wrapping the Python code in a web-based Graphical User Interface, allowing non-technical field teams to run scenario analyses on tablets without requiring support from data scientists.

While the use case focuses on nutrient risk (phosphorus and ammonia) in England, the methodology is highly scalable. Geographically, while the models are currently trained on English datasets, the architecture is replicable across the devolved administrations provided compatible input datasets are sourced. Finally, the open nature of the project offers an opportunity for social scaling. By releasing a version with a user interface to the public, water companies could empower citizen science groups to target their own sampling efforts more effectively, supplementing the monitoring network.

8 References

Defra, 2025. *NCEA River Surveillance Network Analysis Ready Data.* [Online]
Available at: https://environment.data.gov.uk/dataset/21763290-f395-4ca1-bcf6-884c2e7548d4
[Accessed 23 February 2026].

Environment Agency, 2025. *Storm overflow assessment framework 2025.* [Online]
Available at: https://www.gov.uk/government/publications/storm-overflow-assessment-framework-2025/storm-overflow-assessment-framework-2025
[Accessed 2 February 2026].

Independent Water Commission, 2025. *Final report,* s.l.: s.n.

SEPA, 2007. *Scotland's WFD aquatic monitoring strategy.* [Online]
Available at:
https://www.sepa.org.uk/media/38220/wfd_aquatic_monitoring_strategy-scotland_river_basin.pdf
[Accessed 5 December 2025].

The Foundation for Water Research, 2018. *Urban Pollution Manual (UPM).* [Online]
Available at: https://fwr.org/publication/urban-pollution-manual-upm/
[Accessed 4 February 2026].

UK Government, 2013. *The Bathing Water Regulations.* [Online]
Available at: https://www.legislation.gov.uk/uksi/2013/1675/contents
[Accessed 5 12 2025].

UK Government, 2017. *The Water Environment (Water Framework Directive) (England and Wales) Regulations 2017.* [Online]
Available at: https://www.legislation.gov.uk/uksi/2017/407/contents
[Accessed 4 02 2026].

UK Government, 2021. *The Environment Act.* [Online]
Available at: https://www.legislation.gov.uk/ukpga/2021/30/section/82
[Accessed 2 February 2026].

UK Government, 2022. *Water industry national environment programme (WINEP) methodology.* [Online]
Available at: https://www.gov.uk/government/publications/developing-the-environmental-resilience-and-flood-risk-actions-for-the-price-review-2024/water-industry-national-environment-programme-winep-methodology
[Accessed 4 February 2026].

Water UK and Environment Agency, 2018. *Storm Overflow Assessment Framework.* [Online]

Available at: https://www.water.org.uk/sites/default/files/wp/2018/12/SOAF.pdf [Accessed 2 February 2026].

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